

Coding & Solution

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology track.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Sakshi-Bhiryani/AI-Knowledge-Retention-Prediction.git
Programming Language(s)	Python 3.10+
Framework(s) Used	Streamlit, Plotly, SQLite3, Google Gemini API Client
Key Features Implemented	• Ebbinghaus memory retention decay curve generation using NumPy/SciPy. • Dynamic AI quiz generation via Google Gemini API.

	• SQLite persistent database storage for user study logs and performance history.
Pending / Incomplete Features	• Interactive retention tracking dashboard rendered with Plotly. • Push notifications / email reminders for scheduled review sessions. • Multi-user role management (Teacher / Student dashboard views).
Setup / Run Instructions	1. Clone the repository: git clone [https://github.com/sakshibharyani/AI-Knowledge-Retention-Predictor.git] 2. Install dependencies: <code>pip install -r requirements.txt</code> 3. Set API key environment variable: <code>export GEMINI_API_KEY="AQ.Ab8RN6KnfLqxbm9mIQgl1WHIBHiHUT6s450M9hS1I11kb0Qavg"</code> 4. Run application: <code>streamlit run app.py</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1.	Code is modular and organized into functions / classes	Yes
2.	Meaningful variable and function names are used	Yes
3.	Code includes comments / documentation where necessary	Yes
4.	Error handling is implemented for critical operations	Yes
5.	The application runs without critical errors	Yes
6.	Code is committed to a version control repository	Yes

Additional Notes/Comments:

The application successfully meets all core functional requirements by connecting mathematical retention models with generative AI quiz creation. The project structure enforces clean separation of concerns, ensuring high maintainability and ease of future extension.